On the forecasting of record-breaking extremes: benchmarking probabilistic AI weather forecasts

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Abstract

Recent AI weather models achieve impressive average forecast skill, but average scores do not tell us whether they can forecast unprecedented extremes. Motivated by the HEAT colab, and recent evidence that deterministic AI models systematically underpredict record-breaking events, we study whether probabilistic AI forecasts better extrapolate beyond historical records. We first explain the deterministic record-breaking benchmark for heat, cold and wind records, and then extend it to ensemble forecasts. This requires evaluating not only point summaries, but also record probabilities, precision–recall curves, Brier scores, reliability diagrams, CRPS , ... The first results seem to suggest that probabilistic AI changes the picture. Google Deepmind's GenCast performs particularly well for heat records: it assigns probability mass beyond the historical record, has lower record-conditioned CRPS and RMSE than the physics based ensemble IFS-ENS, and shows much lower out-of-ensemble rates. In light of the HEAT project, this is encouraging: extreme heat events appear to be relatively well extrapolated by GenCast. This does not hold uniformly, however. Cold-record intensity and large wind-record margins remain problematic. Thus, probabilistic AI seems to largely fix the heat-record soft cap, but tail calibration remains variable-dependent and should be tested directly.

1 Introduction

Two weeks ago two important papers came out. On Friday I woke up with the headlines of Copernicus's *European State of the Climate 2025* in the news. The report describes Europe as the fastest-warming continent and emphasizes record heat, marine heatwaves, drought, low snow cover, increased wildfire risk and hydrological extremes as non ignorable features of the current European climate (C3S/ECMWF and World Meteorological Organization, 2026). This immediately sparked some discussion at the BIOSTAT office, especially in the context of the HEAT project, where we assess the utility of data-driven models for predicting extreme heat events.

The out-of-distribution problem in machine learning is a known limitation (Engelke et al., 2026), and it led Francesco to claim: "*I don't think it will ever be possible.*"[1] That same evening, Max, tired of the ever-growing arrogance and acronym soup in the AI bubble, sent me a paper that had literally appeared that same day in *Science Advances*. The paper was by Zhongwei Zhang, Erich Fischer, Jakob Zscheischler and Sebastian Engelke (Zhang et al., 2026); not exactly an anonymous group, with Engelke known for extremes, Zscheischler for compound events, and Fischer for climate-extreme projection studies.

[1] Francesco (BIOSTAT), mathematician, is currently studying the potential of neural networks in the context of chaotic systems.

The paper was a bit of a cold shower. Zhang et al. (2026) show that the deterministic AI weather models, what we may call the *first generation* of AI weather forecasters, are not reliably able to forecast events beyond their training range. More precisely, they show that GraphCast, Pangu-Weather and FuXi systematically underestimate record-breaking events. This is worrying, because these models do look very good under the usual verification metrics. On global (latitude-weighted) average scores, the AI models frequently match or outperform physics-based numerical weather prediction systems. The standard deterministic scores are RMSE and anomaly correlation coefficients (ACC), and benchmarks such as WeatherBench 2 have been very useful in making these comparisons a bit more systematic (Rasp et al., 2024).

The implications are quite severe and are directly related to the S2S dream of predicting heatwaves. Accurate forecasts actually matter most during extreme weather situations: heatwaves, where timely warnings can save lives; severe storms, where infrastructure needs to be protected; and hydrological extremes, where crisis management depends on anticipating impacts before they unfold. In such situations, weather forecasts form the basis for early warning systems and disaster response. Which is critical if we want to actually implement data driven models in early warning systems and/or and compliment them with physics-based models. AI weather models learn patterns from historical datasets and use that knowledge to make new forecasts. This works extremely well as long as future situations resemble what has already been observed. But that is precisely where the problem lies. Record-breaking extremes are, by definition, rare, and sometimes entirely new. It is therefore not enough for a model to be good on average. A model can have excellent RMSE because it predicts ordinary weather very well, while still failing on rare extremes intensity and occurrence - wise. That is precisely the issue: the model may win on average, but lose exactly where society cares most. And this is definitely not new to us. As Cas already showed us in his benchmarking study that selected heatwaves between 2022 and 2024 were often badly captured by AI models. Similarly, several model papers include some kind of "extreme-weather" benchmark, but usually this means selected high-impact cases or moderately extreme thresholds. For example, GraphCast and Pangu-Weather evaluate tropical cyclone tracking as an extreme-weather component (Lam et al., 2023; Bi et al., 2023). Such diagnostics are useful, but they are not the same as asking whether the model can forecast genuinely never-seen-before events.

This is what makes the Zhang paper so nice. They explicitly point out that many previous evaluations of extremes either use relatively moderate thresholds, such as the 95th percentile, or rely on isolated case studies of particular heatwaves, storms or tropical cyclones. Their critique is not that these studies are useless. Rather, the point is that they do not really provide a *systematic* test of unprecedented events, which is most likely how our and (our children's) future in West-Europe will look (C3S/ECMWF and World Meteorological Organization, 2026). The most interesting contribution of Zhang is therefore not only the result, but the clean protocol: define record-breaking events locally, relative to the training period, and then test whether the forecast model can predict them, i.e. it turns it into an extrapolation test.

And the result is clear. On average RMSE, AI models often look excellent. But when the evaluation is restricted to record-breaking heat, cold and wind events, the situation flips. HRES, the physics-based ECMWF model, consistently outperforms GraphCast, Pangu-Weather and FuXi (Chen et al., 2023; Lam et al., 2023; Bi et al., 2023). Even more importantly, the errors are not just random. The AI models systematically *underestimate the intensity and frequency of records.* They underpredict hot records, overpredict cold records, and the error grows with the size of the record exceedance. This looks like a soft cap as they put it: the forecasts are pulled back toward the historical range seen during training.

And Max did not pick this up by heavily browsing Google Scholar upon supporting his claims. The internet literally flooded with blog posts and news articles after the publication. See for example this Bloomberg article, this Noodweer Benelux article by Lander Van Tricht, and this Carbon Brief article. Importantly for us , Engelke tells Carbon Brief that probabilistic models indeed "create several possible future states of the weather" and are therefore more likely to capture record-breaking extremes. He also says that it is important to evaluate this newer generation of models for their ability to forecast weather extremes. In other words, the obvious next question is whether the second generation of probabilistic AI weather models passes the same test.

The discussion of the paper already points in this direction. They acknowledge that, during the review process, new probabilistic AI weather models appeared, and they write that these models likely face similar extrapolation challenges when forecasting out-of-distribution record-breaking events. That's where I picked in, since I had been learning about Zarr files, Google buckets and the WeatherBench 2 platform to work with global weather forecasts with the view to extremes postprocessing , this was a logical exercise. The main difference with deterministic forecasts, is that probabilistic forecasts don't just hit or miss. No, in this case we want to know if these models assign any probability mass beyond the historical record, ie are the observed events (correctly calibrated) inside the ensemble envelope? An ensemble represents uncertainty, but it can still be quite underdispersive in the tail. It can still place too little probability mass beyond historical records. The soft cap may or may not disappear.This requires some other metrics than biases and squared errors as we'll discuss in section 5.

If deterministic AI forecasts fail because they are too conservative near the edge of the training distribution, then an ensemble may or may not solve the problem. You know, probably not, but someone has to check. So why not us - it's important to understand the performance of these models in great detail. It seems low-hanging fruit as I described it, but scientifically meaningful and necessary low-hanging fruit: take the record-breaking benchmark and apply it to the next generation of AI weather forecasts, with ECMWF's pride and joy IFS-ENS as the physical ensemble baseline. Bearing in mind our problem setting and the initial ideas to postprocess ML output using extreme value statistics, we have to know how exactly the latest models perform in the extreme extrapolation context for which EVT was made; and in the meantime getting the hang of benchmarking global weather forecasts. On the other hand, the fruit might be so low we'll better write it quick and not too dirty and put it on the arXiv to ensure pioneer-ity.

In the first part I go through the result of the Zhang paper. Their code is available on Zotero, so I could reproduce all the plots using the provided netCDF files fairly easily. In the second part I'll briefly explain how we assess probabilistic forecasts and how Google's Gencast[2] performs in 2020, since this is the only forecast data that is available online in 0.25° resolution. A more systematic study should be done asap, where we look at ECMWF's ensemble models AIFS-ENS and AIFS-CRPS (Lang et al., 2026) as well as the ArchesWeatherGen model from the Sorbonne university (Couairon et al., 2026) and finally the operational models WeatherNext Gen and WeatherNext 2 by Google Deepmind Alet et al. (2025); but these are all less easily accessible. Fuxi-ENS is being prepared but I don't expect great advancements wrt GenCast...

[2]and later FuXi-ENS in 2018

2 Record-breaking forecasts

Let $x_{s,t}$ denote a weather variable at grid cell s and time t. For a variable x, grid cell s and calendar month m, the historical upper record over the training period is

$$r_{s,m}^{x,\max} = \max_{t \in T_{\text{train}},\, m(t)=m} x_{s,t}. \tag{1}$$

For lower-tail records (such as cold records) the maximum is obviously replaced by a minimum,

$$r_{s,m}^{x,\min} = \min_{t \in T_{\text{train}},\, m(t)=m} x_{s,t}. \tag{2}$$

An upper-tail record event in the test period is then a location–time pair satisfying

$$(s,t) \in R^{x,\max} \quad \Longleftrightarrow \quad x_{s,t} > r_{s,m(t)}^{x,\max}. \tag{3}$$

For cold records, the inequality is reversed. The record set therefore contains events that are unprecedented relative to the training period at that location and time of year. In the terminology of extreme value statistics, the test set focuses on threshold exceedances beyond a local historical maximum or minimum. In the terminology of machine learning, it tests out-of-distribution generalization.

2.1 Record dataset

Zhang et al. (2026) study three types of local records:

- heat records: upper records of 2-m temperature;
- cold records: lower records of 2-m temperature;
- wind records: upper records of 10-m wind speed.

The historical records are computed from ERA5 over 1979–2017, which is the training period of the models. The evaluation uses 00 and 12 UTC initializations and lead times that are multiples of 12 hours. The main analysis evaluates land grid cells[3], excluding Antarctica. For the 2020 ERA5 test year, this construction yields 162,754 heat records, 32,991 cold records and 53,345 wind records over land as is shown in figure 1 (above).

[3]defined by the ECMWF's land-sea mask that defines a grid cell land if it contains more than 50% land

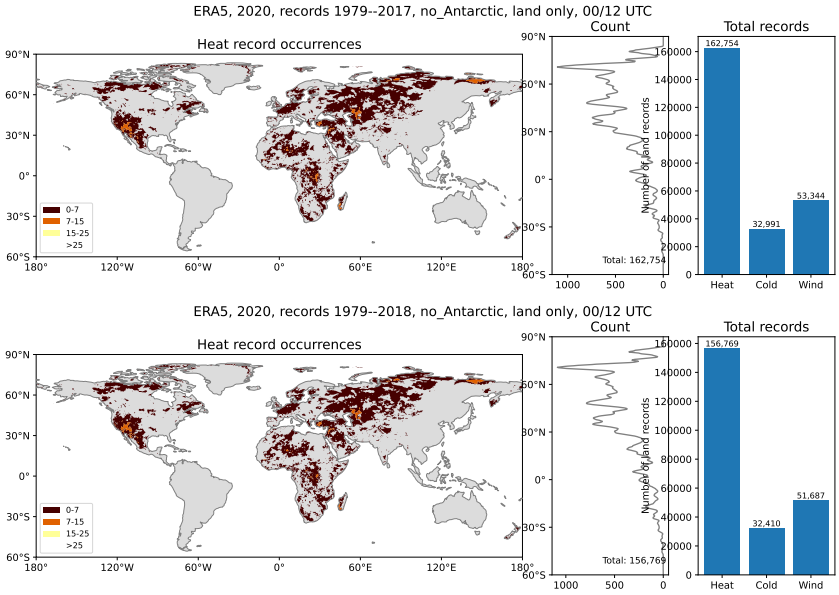


Figure 1: Upper: spatial distribution of record-breaking 2020 heat events, following the record definition of Zhang et al. (2026). Below: update with 2018 records for the later GenCast benchmark, see 3. Difference in counts is negligible.

The benchmark compares GraphCast (Lam et al., 2023), Pangu-Weather (Bi et al., 2023) and FuXi (Chen et al., 2023) with ECMWF's physics-based HRES forecast. Forecasts are accessed through WeatherBench 2 (Rasp et al., 2024). ERA5-trained AI models are verified against ERA5 reanalysis. HRES and operational AI variants are verified against HRES-fc0, the HRES analysis or lead-time-zero forecast, following standard practice.

2.2 Average and extreme skill

For a forecast $\hat{x}^{\tau}_{s,t_0}$ initialized at t_0 with lead time τ, the real[4] verifying value $x_{s,t_0+\tau}$ and a test set I, the latitude-weighted RMSE is

$$\mathrm{RMSE}_I(\tau) = \left\{ \frac{\sum_{(s,t_0)\in I} \omega_s \left(\hat{x}^{\tau}_{s,t_0} - x_{s,t_0+\tau}\right)^2}{\sum_{(s,t_0)\in I} \omega_s} \right\}^{1/2}, \tag{4}$$

where ω_s denotes the grid-cell area weight (since higher latitudes have smaller areas (Zhang et al., 2026)). On all land grid cells and all verifying times, the deterministic AI models perform very well, often matching (for t2m) or even outperforming (for w10) HRES. However, when I is restricted to observed record-breaking events, the ranking flips. HRES has lower RMSE than the deterministic AI models for heat, cold and wind records across almost all lead times. This is illustrated in Figure 2.

[4]Whatever "real" may be for the forecast system as discussed above.

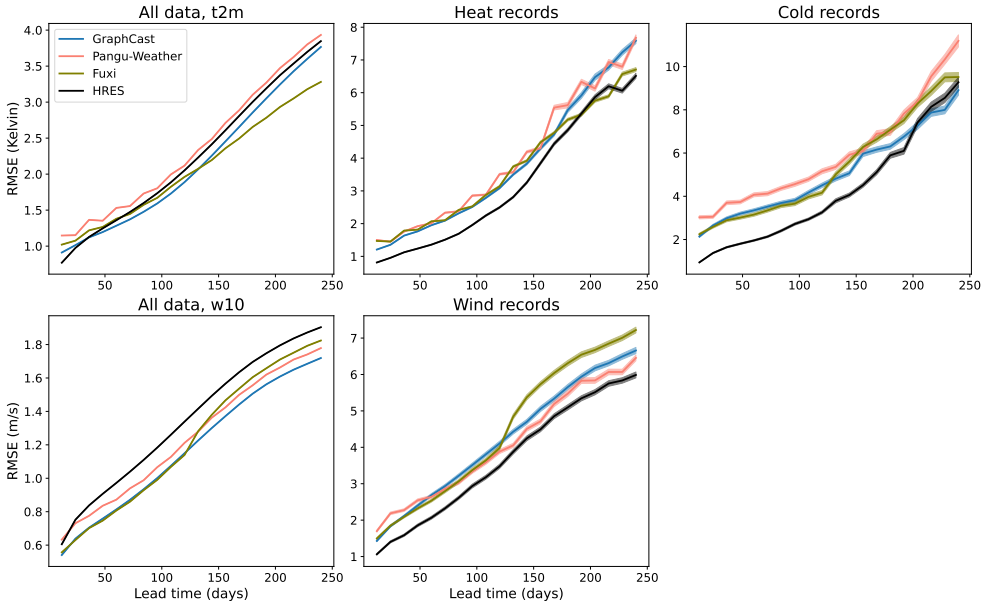


Figure 2: Average RMSE and record-conditioned RMSE in the deterministic benchmark. We see that aggregate skill and record-extreme skill can give different model rankings.

2.3 Intensity errors

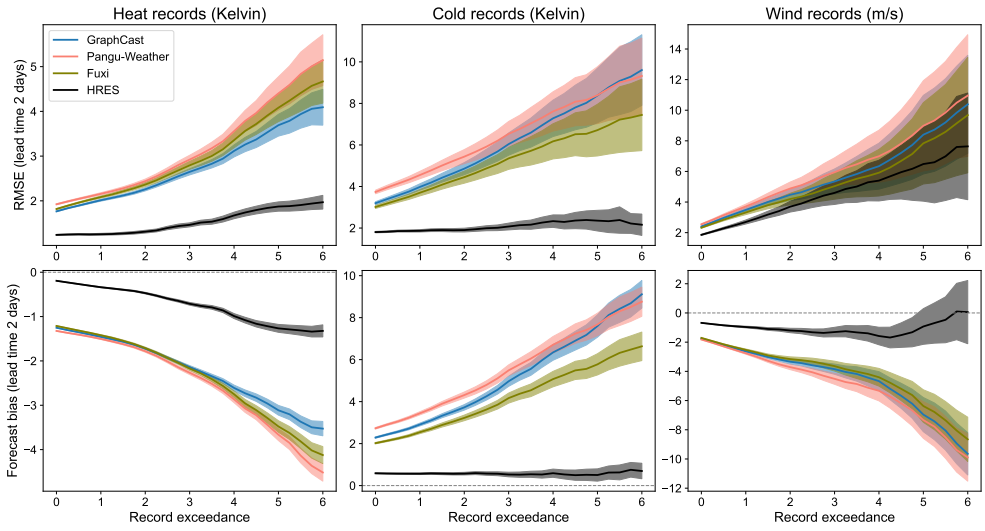


Figure 3: RMSE and forecast bias as a function of the margin.

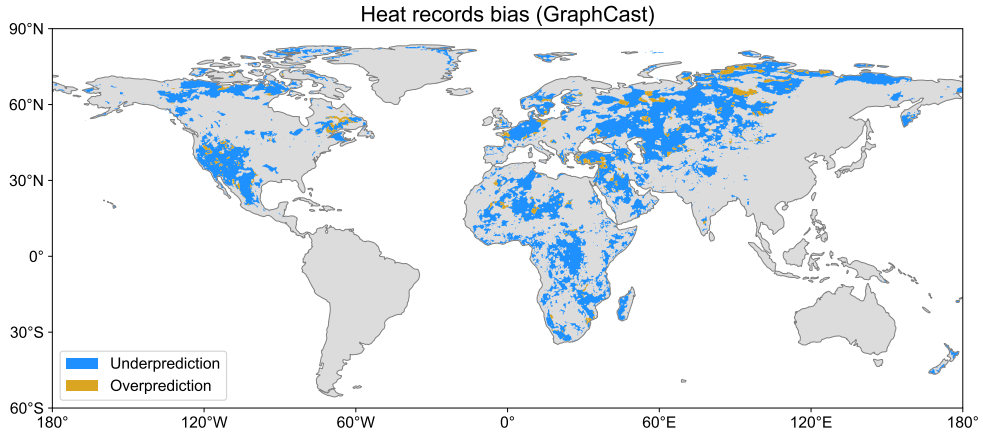


Figure 4: Spatial pattern of GraphCast systematic bias for 2020 heat records at a lead time of two days.

The benchmark further separates random forecast error from systematic bias. The weighted forecast bias over an evaluation set I is

$$\mathrm{FB}_I(\tau) = \frac{\sum_{(s,t_0)\in I} \omega_s \left(\hat{x}^{\tau}_{s,t_0} - x_{s,t_0+\tau}\right)}{\sum_{(s,t_0)\in I} \omega_s}. \tag{5}$$

Define the record margin (for maxima) as

$$e^{\max}_{s,t} = x_{s,t} - r^{x,\max}_{s,m(t)}. \tag{6}$$

When forecast errors are stratified by this margin, the deterministic AI models show an increasingly one-sided error as the record becomes more extreme. Figure 3 shows this for a fixed lead time of 48 hours. GC, Pangu and Fuxi severely underpredict record intensity. The error grows almost linearly with the record margin, whereas HRES remains more or less balanced. Figure 4 shows this bias for the performance of GraphCast on record-breaking heat extremes. For a model based on physical laws it doesn't matter whether the heat event has a history or doesn't, but for a model based on data that's excellent at interpolating, it certainly does. This suggests that the deterministic AI forecasts are pulled back toward values that resemble the training distribution and have an implicit 'soft cap' near the most extreme values encountered during training.

This soft-cap explanation is further illustrated in Figure 5. A natural way to define out-of-distribution samples is through a convex hull: the smallest polygon containing the training data. Points outside this hull require extrapolation , rather than interpolation. In high-dimensional weather data this is impossible to use directly, so Zhang et al. use a simpler definition: a point is simply treated as out-of-distribution if at least one relevant variable exceeds its local historical range. The green rectangle in panel C is this marginal range (and encompasses the blue convex hull). The orange points fall outside it, because they break the historical temperature or wind record.

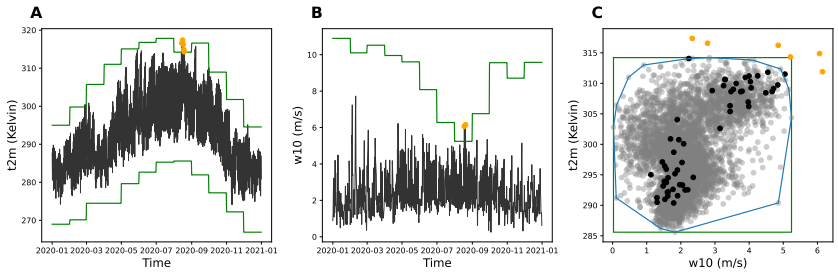


Figure 5: Panels A and B show daily 2020 2-m temperature and 10-m wind speed at one grid cell in California (latitude 37.5°, longitude −121°). The green step functions are the monthly historical records from the training period, and orange points are 2020 observations breaking those records. Panel C shows t2m w10 observations in August: grey points are training data, black points are the 2020 data, the blue polygon is the convex hull of the training data, and the green rectangle is the marginal historical range. Record-breaking out-of-distribution observations lie outside this range.

2.4 Occurrence errors

Apart from intensity errors, we should also ask whether the models can predict the *occurrence* of record-breaking events. For an upper-tail record, the forecast gives the binary indicator

$$\widehat{y}^{\tau}_{s,t_0} = \mathbf{1}\left\{\hat{x}^{\tau}_{s,t_0} > r^{x,\max}_{s,m(t_0+\tau)}\right\}, \tag{7}$$

with the analogous observational indicator

$$y^{\tau}_{s,t_0} = \mathbf{1}\left\{x_{s,t_0+\tau} > r^{x,\max}_{s,m(t_0+\tau)}\right\}. \tag{8}$$

For cold records, the inequality is obviously reversed. We then construct the confusion matrix over an evaluation set I: for example,

$$\mathrm{TP} = \sum_{(s,t_0)\in I} \mathbf{1}\left\{\widehat{y}^{\tau}_{s,t_0} = 1,\, y^{\tau}_{s,t_0} = 1\right\}, \tag{9}$$

and analogously for FP, FN and TN.

We then assess the classification performance by two main quantities,

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} = \mathrm{TPR}, \qquad \mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}. \tag{10}$$

Recall asks "of all records that occurred, how many did the model catch?" whereas the precision asks "of all predicted records, how many were actually records?" In highly imbalanced setting like this one it's more natural to look at these quantities than the more familiar[5] ROC curves where we would plot the recall (or sensitivity) against the false positive rate (and report an AUC)

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}. \tag{11}$$

The number of true negatives is simply enormous, so FPR can be tiny even when the absolute number of false alarms is large.[6] The model can have high accuracy by predicting "no record"

[5] For Stijn and me at least.

[6] As a back-of-the-envelope calculation for 2020 heat records: on the no-Antarctic 0.25° grid there are approximately $601 \times 1440 \times 366 \times 2 = 633{,}502{,}080$ grid-cell/time pairs at 00 and 12 UTC. With 162,754 observed

almost everywhere, while being completely useless for early warning. Precision–recall curves thus ignore the huge number of these easy true negatives and focus on the rare positive class.

For a deterministic forecast, the rule in (7) gives only one point. To obtain a curve, Zhang [7] varies the effective decision boundary by introducing a gain parameter,

$$\widehat{x}^{\tau,g}_{s,t_0} = \widehat{x}^{\tau}_{s,t_0} + g\widehat{\sigma}^{\tau}_{s,m(t_0+\tau)}. \tag{12}$$

In other words, we predict a record whenever

$$\widehat{x}^{\tau}_{s,t_0} > r^{x,max}_{s,m} - g\widehat{\sigma}^{\tau}_{s,m}. \tag{13}$$

Positive g makes the model more aggressive in predicting upper-tail records, increasing recall but also false alarms. Negative g makes the model more conservative, usually increasing precision but missing more records. Varying g traces the precision–recall curve. In the probabilistic case, this will be more natural because we can vary a probability cutoff directly; see Section 4.

Figure 6 shows the difference between the number of forecast records and observed records. The AI models consistently predict too few records, especially for heat extremes. In classification terms, they create many false negatives and seldom predict the occurrence of an OOD event as a consequence of the same soft cap we discussed in the intensity section.

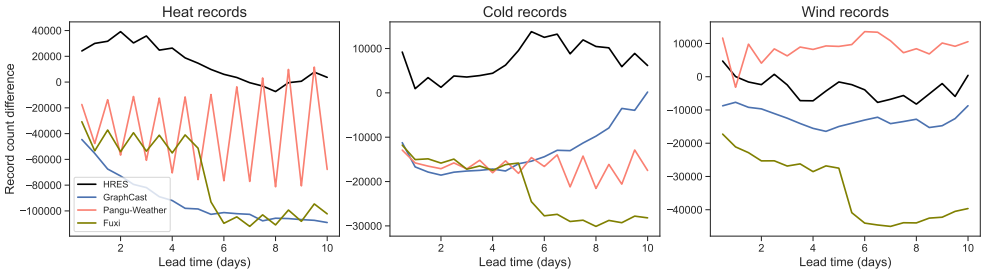


Figure 6: Difference between forecast and observed record counts. Negative values mean that the model predicts too few records.

Figure 7 shows the precision–recall curves. Curves closer to the upper-right corner are better: they combine high recall with high precision. HRES has better curves than GraphCast across heat, cold and wind records, meaning that it catches more records for the same false-alarm rate.

heat records, the prevalence is only $162{,}754/633{,}502{,}080 \approx 2.6 \times 10^{-4}$, so more than 99.97% of the cases are non-records. If a model catches 90% of the heat records, then TP $\approx$ 146,479 and FN $\approx$ 16,275. If it also produces roughly the same order of false alarms, say FP = 146,479, then

$$\text{TPR} = \frac{146{,}479}{162{,}754} \approx 0.90, \qquad \text{FPR} \approx \frac{146{,}479}{633{,}502{,}080 - 162{,}754} \approx 2.3 \times 10^{-4}.$$

The ROC point would therefore look excellent: high recall and almost zero false positive rate. But the precision would only be

$$\text{Precision} = \frac{146{,}479}{146{,}479 + 146{,}479} = 0.50,$$

meaning that half of the record alarms are false, as you'd expect. This would be a useless alarm even though the false positive rate is so tiny.

[7] A similar threshold-variation idea is also used in Lam et al. (2023).

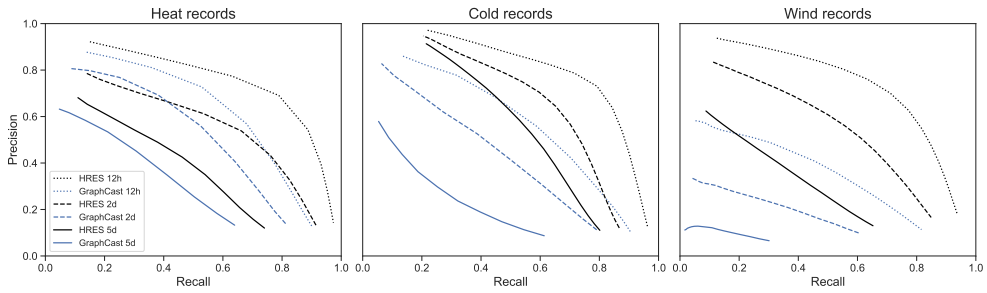


Figure 7: Precision–recall curves for deterministic record occurrence.

Finally, Figure 8 gives correlations between binary record indicators. Higher correlation with the reference means better agreement in the location and timing of record occurrence. HRES is more strongly correlated with its reference than the AI models are with ERA5. The AI models are also positively correlated with each other, which shared successes, but more importantly, shared failures.

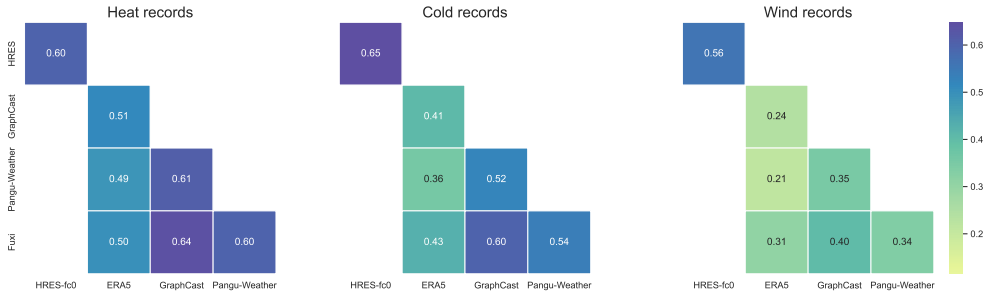


Figure 8: Correlations between binary record indicators for forecasts and verification data.

The occurrence result therefore matches the intensity result: deterministic AI models underestimate the magnitude of records and predict too few of them. Both are consistent with a systematic extrapolation problem.

2.5 The forecaster's dilemma

Conditioning forecast evaluation on observed extremes can create the *forecaster's dilemma*: a model that forecasts too many extremes can be favoured when scoring only cases that were extreme in the observations (Lerch et al., 2017; Holzmann and Eulert, 2014). The study addresses this in two manners. First, it reports performance on the full data set as well as on the record-conditioned subset. And secondly, it also considers 'forecast-conditioned' record sets, where the evaluation set is defined by forecasts exceeding the historical record rather than the observations. The main conclusion does remain the same though: HRES outperforms the AI models on records.

3 Probabilistic AI weather forecasts

A deterministic forecast provides a single future state,

$$\hat{x}^{\tau}_{s,t_0}, \tag{14}$$

for variable x at grid cell s, initialization time t_0 and lead time τ. An ensemble forecast instead provides K possible future states,

$$\left\{\hat{x}^{\tau,1}_{s,t_0}, \ldots, \hat{x}^{\tau,K}_{s,t_0}\right\}. \tag{15}$$

The ensemble can always be reduced to a point forecast, for example by taking the ensemble mean,

$$\bar{x}^{\tau}_{s,t_0} = \frac{1}{K}\sum_{k=1}^{K} \hat{x}^{\tau,k}_{s,t_0}. \tag{16}$$

We can therefore repeat the record benchmark using the ensemble mean or median as a deterministic forecast; these results are shown in Subsection 3.1. However, this reduction discards the main object of interest in a probabilistic forecast: the predictive distribution itself.

In the deterministic case, the soft cap appeared as a systematic underprediction of record intensity and as too few predicted record occurrences. In the probabilistic case, the analogous failure would be an ensemble distribution that remains too concentrated within the historical range. Observed records may then lie systematically outside the ensemble envelope, or occur more often than expected from the forecast probabilities. Section 4 treats record occurrence as a probabilistic classification problem, whereas Section 5 considers record intensity and the corresponding probabilistic skill scores.

We consider two probabilistic AI weather models. The first is *GenCast*, a conditional diffusion model producing stochastic 15-day forecasts at 12-hourly steps and 0.25° resolution (Price et al., 2025). It models the distribution of the next atmospheric state conditional on the two previous states,

$$P(X_{t+1} \mid X_t, X_{t-1}), \tag{17}$$

and generates trajectories autoregressively. At each step, the model starts from random noise and iteratively denoises it into an plausible atmospheric state. Different random noise draws will then produce different weather trajectories, and hence an ensemble.

GenCast is related to the GraphCast architecture (Lam et al., 2023) in the sense that it maps gridded fields to a sphere, processes them there, and maps the result back to the grid. The difference though is that GenCast replaces the deterministic next-state prediction by diffusion-based sampling.[8] The WeatherBench 2 archive contains 50-member GenCast forecasts for 2020 at 0.25° resolution. Since GenCast was trained using data through 2018, we update the ERA5 record dataset to 1979–2018 for this comparison.[9] This updated record set is shown in Figure 1 below.

The second model is *FuXi-ENS*, which also produces global ensemble forecasts at 0.25° resolution up to 15 days ahead, but at 6-hourly rather than 12-hourly steps (Zhong et al., 2025). This is the only other publicly available ensemble model at the same fine resolution that I could find. The available FuXi-ENS archive for 2018[10] contains 48 members and is available

[8] In other words, GenCast is not just GraphCast with noisy initial conditions! In the paper they explicitly compare it to GenCast-Perturbed, a perturbed deterministic model, and show that this produces blurry members, whereas GenCast's diffusion samples remain sharp and spectrally realistic whereas the ensemble *mean* is blurry.

[9] A detail I initially missed, but luckily it did not change any of the main conclusions, meaning that most 2020 records were records with respect to both the 1979–2017 and 1979–2018 period.

[10] This one *is* trained until 2017.

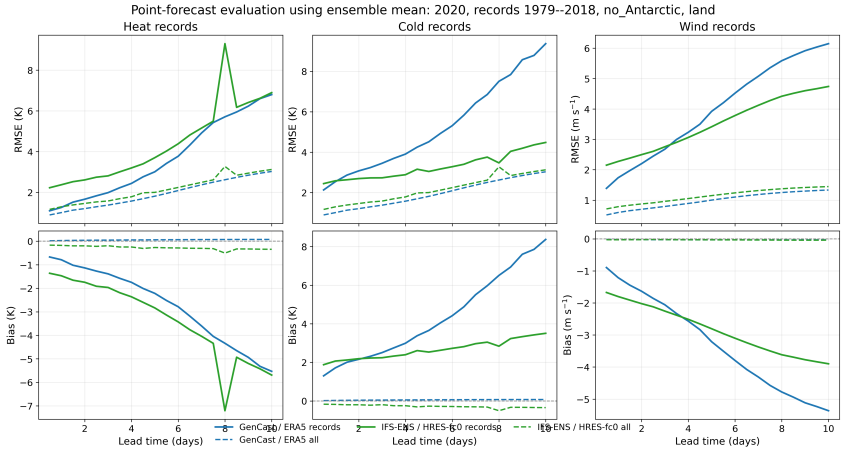


Figure 9: Point-forecast evaluation of the ensemble mean for GenCast and IFS-ENS in 2020, using ERA5 records from 1979–2018 and land grid cells outside Antarctica. Solid lines show scores conditioned on observed record-breaking events; dashed lines show scores over all verification cases. The upper row gives RMSE and the lower row gives forecast bias.

on Hugging Face. However, this is still being processed at the moment of writing. We should eventually extend the setup to additional systems such as AIFS-ENS/CRPS, WeatherNext (2) and ArchesWeatherGen, as discussed in the introduction. Since I would not let you guys wait any longer, I'll await the available results of the Fuxi-ENS record benchmark and later on finish it. We thoroughly discuss the GenCast benchmark though, which has already some surprising results. The HPC meltdown already caused enough delay.

The physical reference system here is ECMWF's ensemble forecasting system: IFS-ENS. IFS-ENS is similarly available through WeatherBench 2 for 2018-2022. We start by considering the point summaries of the ensemble predictions.

3.1 Point summaries of ensemble forecasts

The RMSE and forecast bias in (4) and (5) can simply be computed for the ensemble mean (16), as if it were a deterministic forecast. This gives a first, deliberately reduced, view of the probabilistic forecasts: if the ensemble is collapsed to some central point trajectory, does it still show the same degradation in the record-breaking context? The corresponding median evaluations are qualitatively similar and are shown in Figure 17 in the appendix.

Figure 9 shows that the answer, for GenCast and IFS-ENS in 2020, depends strongly on the variable. Over all verification cases, both systems have naturally small biases and much lower RMSE than on record-conditioned cases. As before: good aggregate skills does not imply good behaviour on unprecedented extremes. For heat records, both ensemble means are too cold during observed record events. The negative bias becomes progressively more pronounced with lead time. Nevertheless, the GenCast point forecast is less biased and has lower RMSE than IFS-ENS over all lead times whcih already seems to suggest that GenCast is actually quite able to extrapolate beyond its training domain for extreme heat. Some anomalous spikes in temperature around lead time 8 days are related to some missing IFS-ENS files and are discussed in Appendix A.For cold records, the ranking reverses. A positive bias means that the forecast is too warm during observed cold extremes. GenCast shows a large positive bias that increases rapidly with lead time, and its RMSE grows faster than that of IFS-ENS, apart from the shortest lead time. Similarly, for wind records, GenCast performs slightly better at the shortest lead times, but from around day 4 onward both its RMSE and negative bias exceed those of IFS-ENS.

Thus, at the level of the ensemble mean, the record-extreme problem is not removed uniformly: GenCast looks favourable for heat records, but still shows strong degradation for cold and wind records.

However, this should be interpreted cautiously. The ensemble mean is indeed convenient because it allows a direct comparison with the deterministic benchmark, but it is not the object that probabilistic models are designed to optimize. As emphasized by Price et al. (2025), ensemble means tend to be spatially smoother, or "blurrier", than individual ensemble members. The individual members of GenCast and IFS-ENS are intended to represent sharp, physically plausible weather states; the mean is only an average over possible futures. Collapsing the ensemble to its mean therefore gives an unfair test: it asks only about central tendencies whereas the whole idea of an ensemble forecast is to acknowledge that the weather cannot be collapsed to the mean.

Although the ensemble mean may be biased, the ensemble may still assign appropriate probability mass to the observed record. We therefore next evaluate the full predictive distribution through record-exceedance probabilities, expected record counts, calibration diagnostics and proper probabilistic scores.

4 Occurrence errors

We start with the occurrence problem. As in Section 2.4, the question here is not how intense a record-breaking event is, but whether the forecast system assigns probability to the event occurring at all. Does it know that an out-of-training event will happen? For an upper-tail record, the ensemble probability of breaking the historical record is

$$\hat{p}^{\tau}_{s,t_0} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\left\{\hat{x}^{\tau,k}_{s,t_0} > r^{x,\max}_{s,m(t_0+\tau)}\right\}. \tag{18}$$

The corresponding observed record indicator is

$$y^{\tau}_{s,t_0} = \mathbf{1}\left\{x_{s,t_0+\tau} > r^{x,\max}_{s,m(t_0+\tau)}\right\}. \tag{19}$$

For cold records, both inequalities are reversed. In other words, the record-occurrence problem becomes a rare-event probabilistic classification problem. For every grid cell and time, the forecast gives us a probability $\hat{p}^{\tau}_{s,t_0}$, whereas the event is of course binary. In Section 4.3, we delve deeper into how we can use these probabilities in assessing the goodness-of-forecast.

4.1 Record counts

As before, we can look at the (expected) number of record-breaking events. For an evaluation set I, we define

$$\widehat{N}^{\tau} = \sum_{(s,t_0)\in I} \omega_s \hat{p}^{\tau}_{s,t_0}, \qquad N^{\tau} = \sum_{(s,t_0)\in I} \omega_s y^{\tau}_{s,t_0}, \tag{20}$$

where ω_s denotes again the latitude weight. The difference

$$\widehat{N}^{\tau} - N^{\tau} \tag{21}$$

measures whether the ensemble predicts too many or too few records in expectation. So unlike for the deterministic occurrence problem in Section 4 where we counted hard threshold crossings by a single forecast, we now sum the ensemble probabilities.

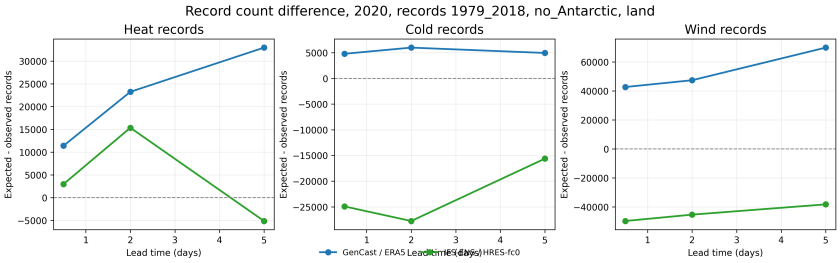


Figure 10: Difference between the expected and observed number of record-breaking events for GenCast and IFS-ENS. Positive values mean that the ensemble predicts too many records in expectation; negative values mean that it predicts too few.

Figure 10 shows that the frequency bias similarly depends on the variable considered - although trends are quite hard to pick up, since I only computed occurrence evaluations for 3 lead times (12h, 2 days, 5 days). GenCast seems to predict too many heat records, and this overprediction increases with lead time whereas IFS-ENS is closer to zero for heat records, although it changes from mild overprediction at short lead times to underprediction at five days. For cold records, GenCast predicts slightly too many records, whereas IFS-ENS predicts far too few. For wind records, the two systems have opposite biases: GenCast predicts again too many wind records, while IFS-ENS predicts too few.

In the benchmark of Zhang, the deterministic AI models tended to predict too few records. Here something else seems to be going on. For GenCast, the occurrence probabilities often lead to *too many* expected records and for IFS-ENS, the sign depends on the variable: heat is roughly balanced, but cold and wind records are underpredicted in total count. However I think count comparisons may be misleading because a model can predict the right number of records in the wrong locations and times... We therefore next evaluate discrimination through the precision–recall curves.

4.2 Precision–recall curves

For a probabilistic forecast, a record alarm can simply be obtained by thresholding the forecast probability, instead of ad hoc varying the forecast via a gain parameter as in (12). For a probability cutoff c, define

$$\widehat{y}^{\tau}_{s,t_0}(c) = \mathbf{1}\left\{\hat{p}^{\tau}_{s,t_0} > c\right\}. \tag{22}$$

Low values of c will produce many alarms, giving high recall but also many false positives, and high values of c will produce very selective alarms, giving higher precision but lower recall. curve.

Figure 11 shows that GenCast has strong occurrence discrimination for heat *and* cold records. For heat records, GenCast lies above IFS-ENS at 12h and 48h over almost the full recall range. At 120h the difference becomes smaller, but GenCast remains competitive. We can already say that the occurrence performance flips upon comparing probabilistic models. For cold records, GenCast's precision–recall curve similarly dominate IFS-ENS at all three lead times. So although the ensemble mean (and median) of GenCast showed strong cold-record intensity errors (bigger than IFS-ENS), the occurrence probabilities identify cold-record cases quite well, and even better than IFS-ENS.

Now, wind records behave somewhat differently. At 12h, GenCast has the better precision–recall curve. By 48h, however, IFS-ENS seems to be better in the high-precision, low-recall part of the curve, which is often the most relevant region for warning applications. At 120h it becomes even stronger: IFS-ENS gives higher precision for useful alarm thresholds, whereas GenCast's

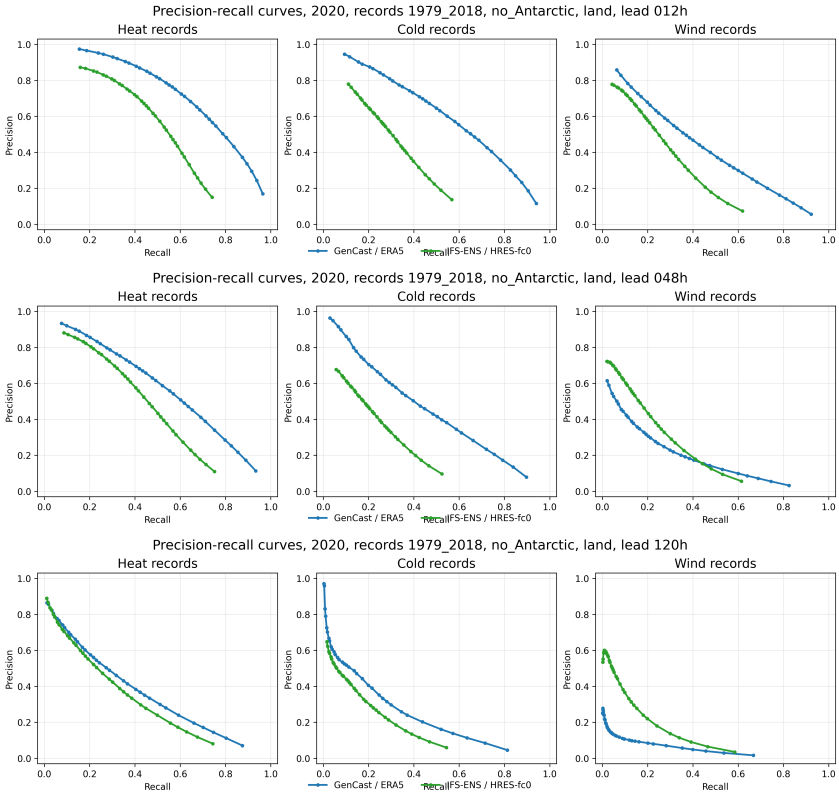


Figure 11: Precision–recall curves for probabilistic record occurrence at lead times 12h, 48h and 120h. Curves closer to the upper-right corner indicate better discrimination .

wind-record discrimination deteriorates substantially. Thus, GenCast is clearly favourable for heat and cold record occurrence, while wind-record occurrence remains more difficult and more favourable to IFS-ENS, especially at longer lead times.

4.3 Brier score

Precision–recall curves basically evaluate whether the cases assigned the highest record probabilities are indeed the cases where records occur. In other words, after ranking all grid-cell/time cases by their predicted probability $\hat{p}^{\tau}_{s,t_0}$, to the top-ranked cases contain many true records and few false alarms ? So if we would use the forecast as an (early) warning system and pick a decision threshold c, do the predicted record alarms correspond to the actual records?

However, precision–recall curves do not check whether the numerical probabilities themselves are calibrated. What I mean by that is that a model may *rank* these cases correctly, but still assign probabilities that are too large or too small. We therefore also evaluate the forecasts directly as probability forecasts. For this purpose we introduce the well-known Brier score (Brier, 1950), a standard scoring rule for (binary) probabilistic forecasts. For a selected set of verification cases I, the latitude-weighted Brier score is

$$\mathrm{BS}_I(\tau) = \frac{\sum_{(s,t_0)\in I} \omega_s \left(\hat{p}^{\tau}_{s,t_0} - y^{\tau}_{s,t_0}\right)^2}{\sum_{(s,t_0)\in I} \omega_s}. \tag{23}$$

For example, if an ensemble assigns probability 0.8 to a record and the record occurs, the squared error is only $(0.8-1)^2 = 0.04$. If it assigns probability 0.05 to the same event, the squared error becomes $(0.05-1)^2 = 0.9025$, so we want it to be as low as possible. The Brier score averages

these probability errors over all record events, and will thus reward ensembles that assign high probability to records that occur and low probability to records that do not occur (Murphy, 1973).

In Section 5, we build on this concept to assess the intensity upon exceeding (i.e. whether there is sufficient probability near the observed record intensity).

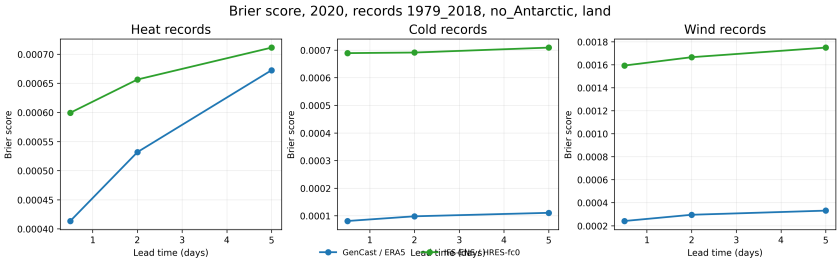


Figure 12: Latitude-weighted Brier scores for probabilistic record occurrence. Lower is better.

Figure 12 shows that GenCast has lower Brier scores than IFS-ENS for all three record types and all displayed lead times. The difference is particularly large for cold and wind records. This confirms that, as a probabilistic forecast of record occurrence, GenCast performs better - for all variables and lead times...

The wind result should nevertheless be interpreted together with the precision–recall and reliability diagnostics. Although GenCast has a lower Brier score for wind occurrence, we saw that its precision–recall curves (and reliability diagrams) deteriorate at longer lead times. This means that the Brier score does not by itself reveal everything. We therefore also inspect calibration or *reliability* directly.

4.4 Reliability

The classical Brier score decomposition separates the score into reliability, resolution and uncertainty (Murphy, 1973):

$$\mathrm{BS} = \mathrm{REL} - \mathrm{RES} + \mathrm{UNC}. \tag{24}$$

Reliability measures calibration error: if the model assigns probability p, does the event indeed occur with frequency p? Resolution measures whether the forecast can separate low-risk from high-risk cases. Finally,

$$\mathrm{UNC} = \bar{y}(1 - \bar{y}) \tag{25}$$

depends only on the climatological event frequency $\bar{y}$. Here I focus on reliability, because it gives the most direct visual check of whether the predicted record probabilities are meaningful. Still, the full decomposition may be worth studying more carefully later, since it could help separate different probabilistic failure modes: bad calibration, poor discrimination, or simply the intrinsic rarity of records. Looking forward to the insights of our old hand in classification, Thomas!

Reliability diagrams bin forecasts according to their predicted probability. We use the bins

$$[0, 0.001),\ [0.001, 0.01),\ [0.01, 0.02),\ [0.02, 0.05),\ [0.05, 0.10), \tag{26}$$

$$[0.10, 0.20),\ [0.20, 0.40),\ [0.40, 0.60),\ [0.60, 0.80),\ [0.80, 1]. \tag{27}$$

They are deliberately finer near zero, because records are rare and the low-probability region is where a probabilistic soft cap should show up. For a 50-member ensemble such as GenCast, the possible probabilities are discrete, namely $0, 1/50, 2/50, \ldots, 1$. Within each bin b, we then

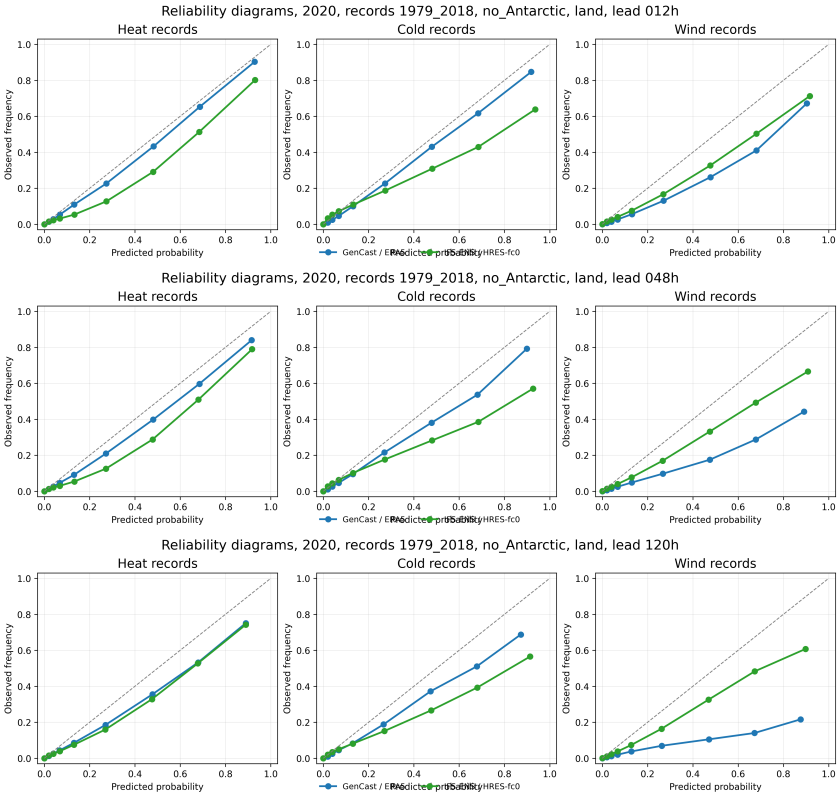


Figure 13: Reliability diagrams for probabilistic record occurrence at lead times 12h, 48h and 120h. The diagonal indicates perfect calibration. Curves below the diagonal overforecast record probabilities; curves above the diagonal underforecast them.

compute the mean predicted probability $\bar{p}_b$ and the empirical record frequency $\bar{y}_b$, and plot $\bar{y}_b$ against $\bar{p}_b$.

The diagonal corresponds to perfect reliability. Points below the diagonal mean overforecasting: records occur less often than predicted. Points above the diagonal mean underforecasting.

As with the record counts, figure 13 is not quite what I expected. Given the deterministic soft-cap result, one might expect the ensembles to underforecast records. Instead, many curves lie below the diagonal, meaning that (conditional on a nonzero forecast probability, see later) records occur less often than predicted. In other words, the main signal in these reliability diagrams is often overforecasting rather than underforecasting. GenCast is generally closer to the diagonal than IFS-ENS for heat and cold records, especially at shorter lead times. Wind is the main exception: at 48h and especially 120h, GenCast lies far below the diagonal for moderate and high forecast probabilities, meaning that it assigns large wind-record probabilities too often.

Since I was suspicious about this result I focused even more on the $\hat{p} = 0$ case, since this is closest to the soft cap story. The first bin of the reliability diagram estimates

$$\Pr(y = 1 \mid \hat{p} = 0), \tag{28}$$

but as we discussed in the introduction of precision recall, record-breaking events are extremely rare. This quantity can thus be tiny even when many observed records are missed with $\hat{p} = 0$. So let's look at the reverse conditional probability,

$$\Pr(\hat{p} = 0 \mid y = 1), \tag{29}$$

Table 1: Zero-probability miss rate, $\Pr(\hat{p} = 0 \mid y = 1)$, over land grid cells outside Antarctica.

Record type	Model	12h	48h	120h
Heat	GenCast	**0.041**	**0.076**	**0.129**
Heat	IFS-ENS	0.219	0.214	0.228
Cold	GenCast	**0.075**	**0.142**	**0.221**
Cold	IFS-ENS	0.471	0.520	0.497
Wind	GenCast	**0.091**	**0.199**	**0.347**
Wind	IFS-ENS	0.368	0.383	0.414

i.e. the fraction of observed records for which no ensemble member crossed the historical record. This is exactly the false negative rate at $c = 0$, since at this threshold we predict a record whenever *at least one member* exceeds the historical record:

$$\Pr(\hat{p} = 0 \mid y = 1) = 1 - \text{Recall}(c = 0). \tag{30}$$

Table 1 shows the results for both models over all lead times. The reliability diagrams suggest overforecasting in many nonzero probability bins, but at the same time a substantial fraction of observed records still receive zero probability. GenCast does much better than IFS-ENS: for heat records it misses only about 4% at 12h and 13% at 120h, compared with about 22% for IFS-ENS. For cold records the difference is even larger, with IFS-ENS assigning zero probability to roughly half of the observed records. Wind remains the most difficult case. GenCast still has fewer zero-probability misses than IFS-ENS, but by 120h it assigns zero probability to about one third of observed wind records.

So let's say the reliability story has two sides. In the nonzero probability bins GenCast seems overconfident, especially for wind at longer lead times. But at the most permissive threshold, $c = 0$, GenCast does seem to be superior : it assigns zero probability to far fewer observed records than IFS-ENS for all variables and lead times. Thus, even when its probabilities are not perfectly calibrated, GenCast is much less likely to completely miss a record-breaking event.

5 Average and extreme (probabilistic) skill

Occurrence only asks wether the ensemble thinks the record can be crossed. But this is not enough. It's possible that the model puts non zero probability right above the threshold, but still place the whole ensemble just barely above the record threshold while the observation might be way more extreme.

5.1 Continuous ranked probability score

We therefore introduce the continuous ranked probability score (CRPS), a standard proper scoring rule for (real-valued) probabilistic forecasts (Gneiting and Raftery, 2007).

For a predictive distribution F^{τ}_{s,t_0} and observation $x_{s,t_0+\tau}$, the CRPS is defined as

$$\text{CRPS}\left(F^{\tau}_{s,t_0}, x_{s,t_0+\tau}\right) = \int_{-\infty}^{\infty} \left(F^{\tau}_{s,t_0}(z) - \mathbf{1}\left\{x_{s,t_0+\tau} \le z\right\}\right)^2 dz. \tag{31}$$

The CRPS can be seen as an integral of Brier scores over all possible thresholds (Hersbach,

2000).

For an ensemble forecast with K members, the empirical CRPS is[11]

$$\widehat{\mathrm{CRPS}}^{\tau}_{s,t_0} = \frac{1}{K}\sum_{k=1}^{K}\left|\hat{x}^{\tau,k}_{s,t_0} - x_{s,t_0+\tau}\right| - \frac{1}{2K^2}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\left|\hat{x}^{\tau,k}_{s,t_0} - \hat{x}^{\tau,\ell}_{s,t_0}\right|. \tag{32}$$

The first term measures how far the ensemble members are from the observation. The second term accounts for the ensemble spread. If the ensemble collapses to one deterministic member, the second term vanishes and the CRPS reduces to the absolute error.

We average this score over I using the same latitude weights as before:

$$\overline{\mathrm{CRPS}}_I(\tau) = \frac{\sum_{(s,t_0)\in I}\omega_s \widehat{\mathrm{CRPS}}^{\tau}_{s,t_0}}{\sum_{(s,t_0)\in I}\omega_s}. \tag{33}$$

A detail that is relevant for implementation: let

$$\hat{x}^{\tau,(1)}_{s,t_0} \le \cdots \le \hat{x}^{\tau,(K)}_{s,t_0}$$

be the sorted ensemble. Then

$$\frac{1}{2K^2}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\left|\hat{x}^{\tau,k}_{s,t_0} - \hat{x}^{\tau,\ell}_{s,t_0}\right| = \frac{1}{K^2}\sum_{k=1}^{K}(2k-K-1)\hat{x}^{\tau,(k)}_{s,t_0}. \tag{34}$$

This avoids forming all pairwise differences, i.e. first sort them and then calculate the variance (Jordan et al., 2019).

Figure 14 gives the probabilistic analogue of the point-summary result in Figure 9. Over all verification cases, the dashed curves show that GenCast has lower CRPS than IFS-ENS for heat, cold and wind. So, on average, GenCast is again the stronger probabilistic system.

But, as before, the record-conditioned evaluation tells a more interesting story. For heat records, GenCast has lower CRPS than IFS-ENS across essentially all lead times. This is the same conclusion as for the ensemble mean: GenCast seems genuinely strong for 2020 heat-record intensity, not only for occurrence.

For cold records the ranking reverses. GenCast's CRPS grows rapidly with lead time and becomes much larger than that of IFS-ENS. This is again the same pattern as in the point-

[11] From the expectation representation

$$\mathrm{CRPS}(F,x) = \mathbb{E}_F|X-x| - \frac{1}{2}\mathbb{E}_F|X-X'|,$$

where X and X' are independent draws from F (Gneiting and Raftery, 2007), we replace F by the empirical ensemble distribution

$$\hat{F}_K = \frac{1}{K}\sum_{k=1}^{K}\delta_{\hat{x}^k},$$

giving

$$\mathbb{E}_{\hat{F}_K}|\hat{X}-x| = \frac{1}{K}\sum_{k=1}^{K}|\hat{x}^k - x|$$

and

$$\mathbb{E}_{\hat{F}_K}|\hat{X}-\hat{X}'| = \frac{1}{K^2}\sum_{k=1}^{K}\sum_{\ell=1}^{K}|\hat{x}^k - \hat{x}^\ell|.$$

Substitution gives Equation (32). Hersbach gives the same ensemble CRPS idea in its ordered-ensemble/CDF form (Hersbach, 2000).

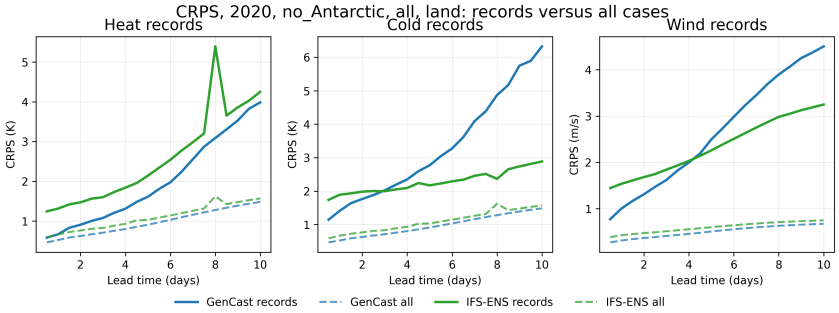


Figure 14: CRPS for GenCast and IFS-ENS in 2020, over land grid cells outside Antarctica. Solid lines show CRPS conditioned on observed record-breaking events; dashed lines show CRPS over all verification cases. Lower is better.

forecast evaluation: GenCast was too warm during observed cold records, and the full ensemble distribution does not fix this error. Wind records are intermediate. GenCast is better at the shortest lead times, but after roughly day 4 IFS-ENS has lower record-conditioned CRPS. This matches the earlier mean-bias result and the occurrence diagnostics: wind remains the most difficult case for GenCast.

Thus the CRPS confirms the main message from the point summaries, but now using the full ensemble distribution. GenCast is clearly favourable for heat records and strong on average, but cold and wind records still expose substantial tail errors. Probabilistic sampling therefore helps, especially for heat, but it does not uniformly remove the record-intensity problem.

5.2 Intensity errors

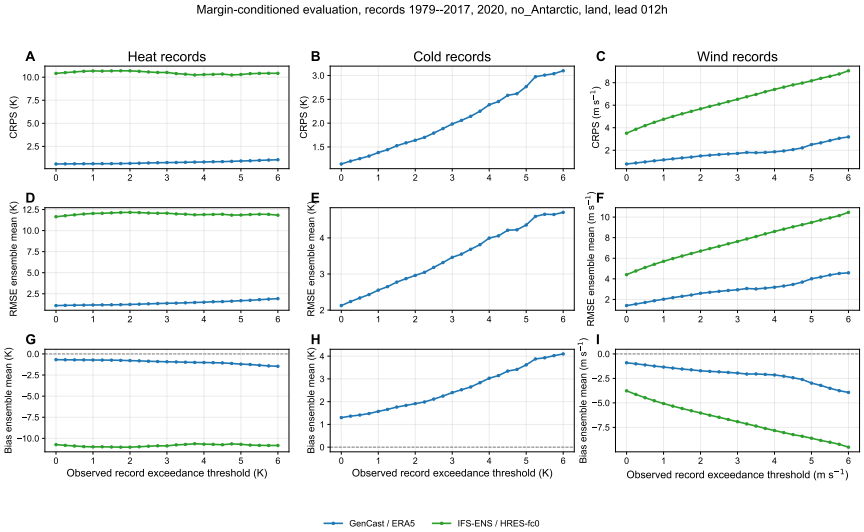
We now follow the spirit of the most memorable diagnostic in Zhang et al. (2026): instead of only averaging over records, we look at forecast performance as a function of the observed record exceedance, or *margin*. In other words, we compute a given diagnostic over all observed records with margin $e > u$ where u is a threshold. We should for a paper definitely add some confidence intervals (Zhang uses an asymptotic argument which I'm a bit confused by currently) since the larger the margin, the fewer the available observations. For each threshold u, we compute the mean CRPS, the RMSE of the ensemble mean, and the bias of the ensemble mean.

From Figure 15 we can learn a lot more than the aggregate CRPS score from the previous section. For heat records, GenCast is clearly better than IFS-ENS at all three lead times and magnitude of the exceedance. Its CRPS and ensemble-mean RMSE are lower, and its bias is much less negative. GenCast seems genuinely strong for heat-record intensity.

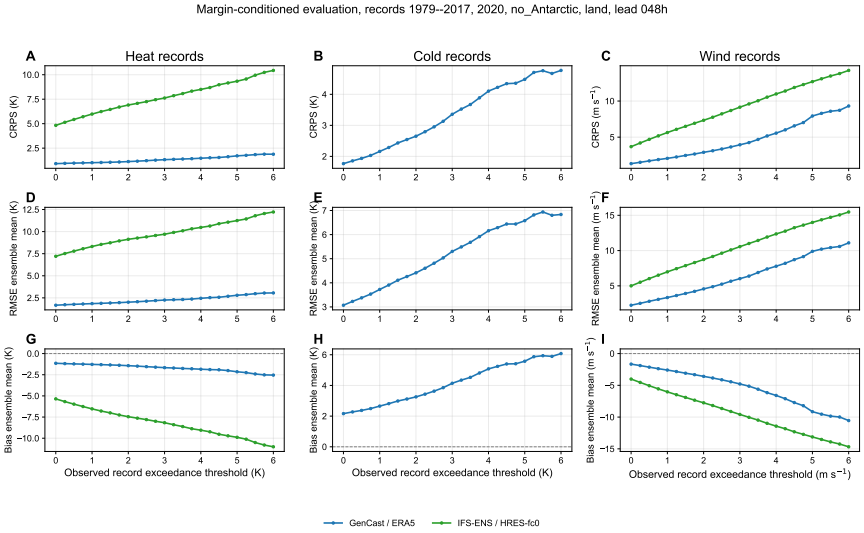
For cold records, the IFS-ENS results are still incomplete, so the comparison is not final. And since the scales are misleading, better be patient before drawing conclusions.

For wind records, GenCast also improves over IFS-ENS. It has lower CRPS, lower RMSE and a less negative bias for all margins and lead times... However, all curves still deteriorate massively as the margin increases, especially at longer lead times. So GenCast is better than IFS-ENS for wind in this figure, but the most extreme wind records are still difficult , for both models.

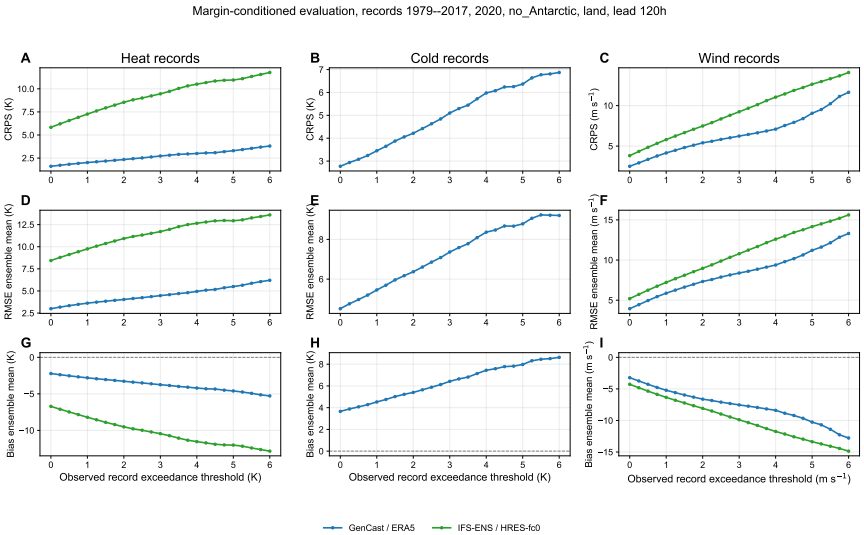
We also look at diagnostics that are more probabilistic. First, we compute the mean forecast record probability $\hat{p}$: among the records with margin $e > u$, how many ensemble members crossed the historical record threshold? And second, we compute the out-of-ensemble (OOE) rate, i.e. how often the observed record lies beyond the entire ensemble. For heat and wind



(a) Lead time 12h.



(b) Lead time 48h.



(c) Lead time 120h.

Figure 15: Intensity vs margin diagnostics (CRPS, RMSE, bias)for 2020 records. Some IFS-ENS cold-record results are still being computed.

records this is

$$o_{s,t}^{\tau} = \mathbf{1}\left\{x_{s,t} > \max_{1\le k\le K} \hat{x}_{s,t-\tau}^{\tau,k}\right\}, \tag{35}$$

with the inequality reversed for cold records. This asks whether the ensemble envelope even contains the observed record.

This is a bit stronger than the false negative rate at $c = 0$ from the occurrence problem (Equations (29) and (30)) which asked whether at least one member crossed the historical record threshold. The OOE rate asks whether at least one member reached the actual record value itself - so it is closer to the intensity problem. As you might have noticed I did not use the familiar rank histograms here. In probabilistic AI weather papers, rank histograms are usually shown over all cases as evidence of ensemble calibration. But after conditioning on observed record-breaking events, the observation is selected from the tail by construction, so we should not expect a flat histogram... What actually matters is the edge of the rank histogram: how often does the observation lie beyond *all* ensemble members? This is exactly the OOE rate.

We show these results in Figure 16. For heat records, GenCast assigns much larger record probabilities than IFS-ENS and has a much lower out-of-ensemble rate. So even when IFS-ENS knows the event to be a record, the ensemble often does not reach the observed value

For cold records, we again only interpret GenCast for now. The record probabilities remain reasonably high, which is consistent with the occurrence results, but the out-of-ensemble rate increases with margin and lead time.

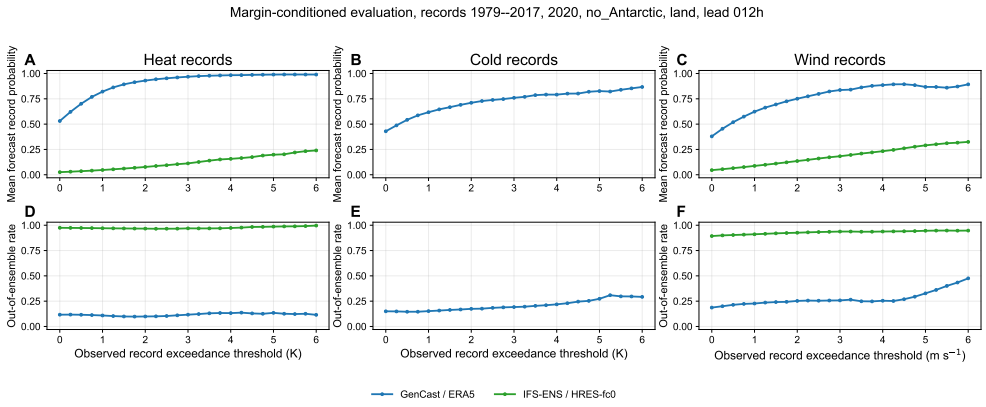
Lastly for wind records, GenCast seems to still have better probabilistic tail behaviour than IFS-ENS, especially at 12h and 48h. It assigns higher record probabilities and has a lower out-of-ensemble rate. But the wind tail is still not solved . At 120h, and especially for large margins, the GenCast OOE rate rises.

6 Discussion

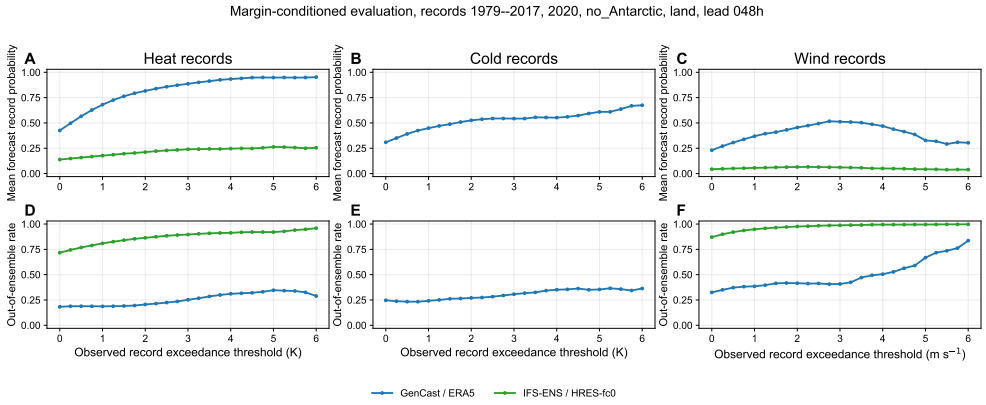
The main conclusion so far is (unfortunately) more interesting than a simple "Google beats IFS" or "IFS beats Google". In the deterministic benchmark of Zhang et al. (2026), the AI models systematically underpredicted record-breaking extremes. In the probabilistic benchmark, the failure modes change. GenCast performs very well for heat records: its ensemble mean has lower RMSE and smaller bias than IFS-ENS, its CRPS is lower, its occurrence probabilities discriminate records well, and the margin-conditioned diagnostics show much less soft-capping. In other words, for 2020 heat records, GenCast does not merely assign *some* probability to record occurrence; it also puts substantially more (calibrated) mass near the observed record intensity. This is in my opinion the strongest and most publishable message of the current analysis.

We saw many times that the picture is not uniform, however. Cold-record occurrence is well captured by GenCast, but cold-record intensity remains problematic: the ensemble is too warm in the lower tail, and the error grows with lead time and margin. Wind records are somewhere in between. GenCast improves over IFS-ENS in several margin-conditioned diagnostics, especially at short lead times, but large-margin wind records still expose underdispersion and increasing out-of-ensemble rates. This variable dependence is scientifically important. It suggests that probabilistic sampling helps with extrapolation, but does not automatically calibrate all physical tails.

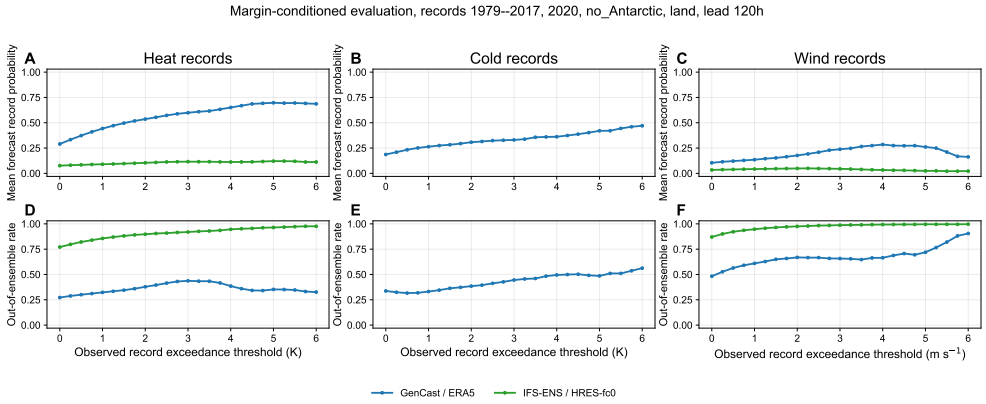
Several steps are still needed before this can become a clean article. The affected IFS-ENS files (see appendix A), should be re- downloaded and computed (and the IFS cold-record margin diagnostics are still awaiting!). On my side, I'll make sure the code is cleaned up maybe with a shared data drive so that the results can be checked without repeatedly downloading from cloud



(a) Lead time 12h.



(b) Lead time 48h.



(c) Lead time 120h.

Figure 16: Margin-conditioned probabilistic diagnostics for 2020 records over land grid cells outside Antarctica. The horizontal axis gives the margin threshold u; each point is computed over records with exceedance $e > u$. Rows show mean forecast record probability, out-of-ensemble rate, and empirical coverage of the central 90% ensemble interval. Some IFS-ENS cold-record results are still being computed.

buckets (which can be slow). FInally we should consider adding similar forecast-conditioned sensitivity analyses to address the forecaster's dilemma, i.e. evaluating cases selected by predicted record probability.

Additional tail scores would strengthen the paper but may not be necessary for the first version. Threshold-weighted CRPS and some quantile scores would be interesting; and maybe to link our research to S. Engelke we can try to incorporate some extreme value diagnostics, to understand why some variables are better captured (extrapolated) outside the training than other. An actual full version could add more years, add FuXi-ENS, AIFS-ENS/AIFS-CRPS, WeatherNext ... regional breakdowns etc.

7 Conclusion

The current results suggest that this analysis can be shaped into a short article, provided the message is kept focused. The story is that deterministic AI models were shown to fail on out-of-training records, while GenCast, as a generative probabilistic model, largely fixes this problem for heat records. This is not true uniformly across variables: cold-record intensity and large wind extremes remain difficult. We should therefore not claim that probabilistic AI has solved record forecasting, definitely not. Rather, it should show that the second generation changes the picture: heat records are surprisingly well captured, occurrence skill is often strong, but tail calibration remains variable-specific and must be tested directly.

A 8 day peak

During the evaluation of IFS, I found missing forecast fields in several Zarr files for 2020, most importantly for 2m temperature at lead time 192h. This explains the anomalous 8-day spike in the temperature diagnostics. So the corresponding scores are based on only a small and unrepresentative subset of the year. The affected files should therefore be checked and regenerated before interpreting the 8 day lead IFS temperature scores.

B Reducing the ensemble to the median

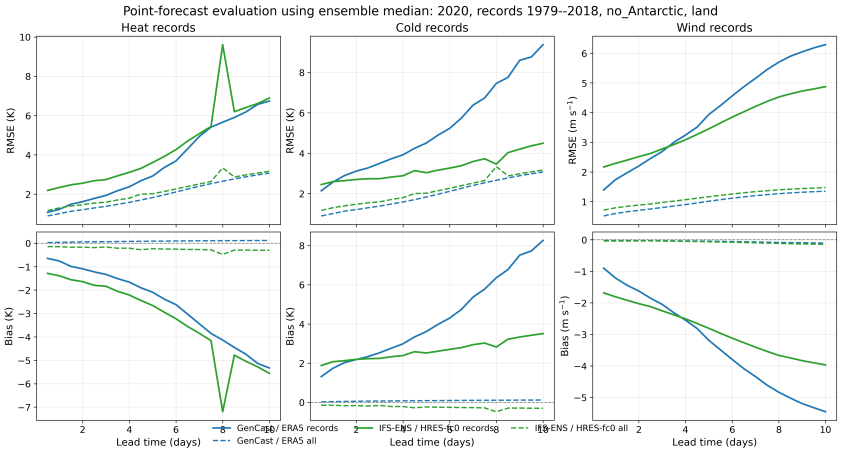


Figure 17: The results for the median collapse.

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